FSD2 LAB • EXPERIMENT 04

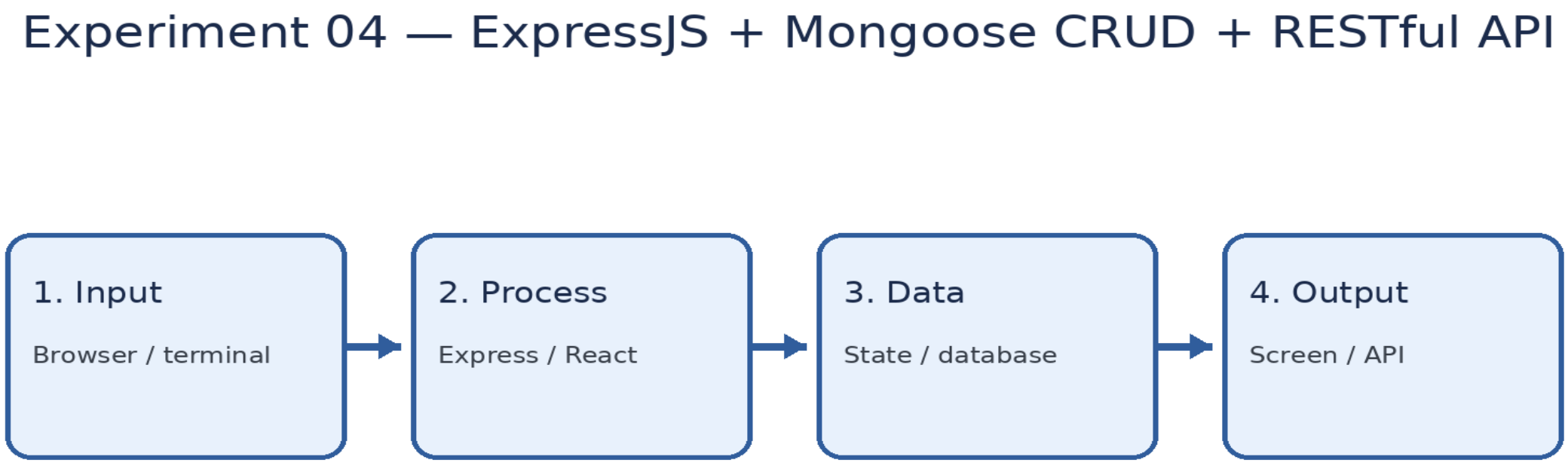
Experiment 04 — ExpressJS + Mongoose CRUD + RESTful API

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- `ECONNREFUSED`: start local MongoDB or supply a correct Atlas URI.
- Invalid URI / Atlas timeout: recheck connection string and Atlas Network Access.
- Invalid product ID: API returns a 400 response.
- Missing dependency: from repository root run `npm install`.

2. Workflow



Workflow: input → processing → data/state → output

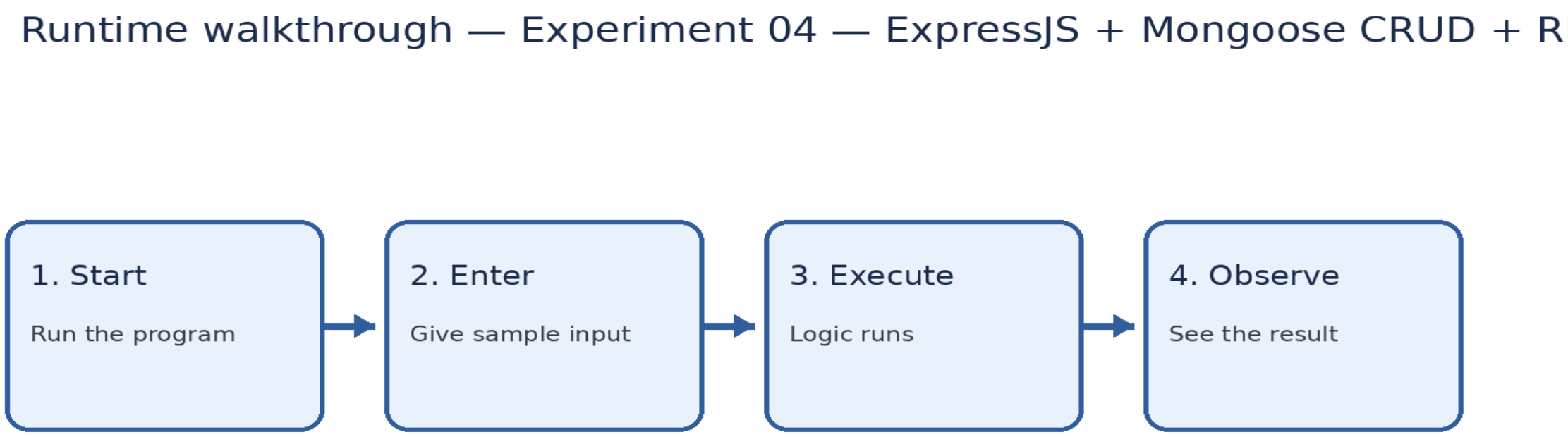
3.1. Code pathway — server.js

```
1 /**
2 * Experiment 04: Express + Mongoose CRUD REST API with browser client.
3 *
4 * Default local database: mongodb://127.0.0.1:27017/fsd2lab
5 * Set MONGO_URI to use MongoDB Atlas.
6 * Run from the repository root:
7 * node 04-express-database-rest-api/server.js
8 */
9 const express = require('express');
10 const mongoose = require('mongoose');
```

3.2. Code pathway — server.js

```
11
12 const app = express();
13 const PORT = Number(process.env.PORT) || 3004;
14 const MONGO_URI = process.env.MONGO_URI || 'mongodb://127.0.0.1:27017/fsd2lab';
15
16 app.use(express.json());
17 app.use(express.urlencoded({ extended: true }));
18
19 const productSchema = new mongoose.Schema({
20 name: { type: String, required: true, trim: true },
```

4. Runtime walkthrough



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?